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Comparison between traditional software development and Low code platform

Feature	Traditional Software Development	Low-Code Platform
Development Approach	Hand-written code	Visual drag-and-drop with minimal coding
Programming Skills	Strong programming knowledge required	Basic to moderate programming knowledge
Development Speed	Slower	Much faster
Customization	Very high	Moderate to high (depends on the platform)
Flexibility	Unlimited	Limited by platform capabilities
Cost	Higher due to longer development time	Lower for many business applications
Maintenance	Developers maintain code manually	Platform handles much of the maintenance
Deployment	Manual configuration and deployment	One-click or automated deployment

Scalability	Highly scalable when designed well	Good, but depends on the platform's limits
Security	Implemented by developers	Many security features are built into the platform
Integration	Custom APIs and code	Pre-built connectors and API support
Testing	Manual and automated testing frameworks	Often includes built-in testing tools
Time to Market	Weeks to months	Days to weeks
Best For	Complex, custom, large-scale applications	Business applications, workflows, and rapid prototyping

Example

Suppose a company needs a **Customer Relationship Management (CRM)** application.

Traditional Development

- Design the UI from scratch.
- Write backend logic.
- Create database schema.
- Develop APIs.
- Implement authentication.
- Test and deploy.

Estimated time: 4–8 months (depending on complexity).

Low-Code Platform

- Use pre-built CRM templates or components.
- Drag and drop forms and dashboards.
- Configure workflows.

- Connect to an existing database.
- Publish the application.

Estimated time: 2–8 weeks (depending on customization).

Advantages of Traditional Development

- Complete control over application design and functionality.
- Ideal for highly customized or complex systems.
- No dependency on a specific low-code vendor.
- Better suited for performance-critical applications.

Advantages of Low-Code Platforms

- Rapid development and deployment.
- Lower development costs.
- Easier for business users to participate in application creation.
- Built-in integrations, security, and monitoring.
- Faster updates and maintenance.

When to Choose Each

Choose Traditional Software Development if:

- You need a highly customized application.
- Performance and scalability are critical.
- You are building operating systems, game engines, financial trading systems, or other complex software.
- You want full control over the technology stack.

Choose a Low-Code Platform if:

- You need to build applications quickly.
- You're creating internal business tools, approval workflows, or dashboards.
- Your team has limited development resources.
- Rapid prototyping or frequent business-driven changes are important.

Summary

Traditional Development	Low-Code Platform
Maximum flexibility and control	Maximum speed and productivity

Requires skilled developers	Accessible to both developers and business users
Longer development cycles	Shorter development cycles
Best for complex, custom software	Best for business applications and process automation

In practice, many organizations use **both** approaches: low-code platforms for internal business applications and workflow automation, and traditional development for customer-facing products or applications that require extensive customization and high performance.

Examples of low-code apps businesses can develop

- Operational cost management apps
- Scheduling apps
- Business expense management apps
- Timesheet apps
- Help desk apps
- Leave request apps
- Service request apps
- Payroll apps
- Grant and contract activity apps
- Approval process apps
- Inventory tracking apps
- Event planning apps

Benefits of low-code development

- **Ease of use:** Drag-and-drop modules and templates take the guesswork out of app development.
- **Reduced IT workload:** By providing IT professionals with an expanded

toolkit, low-code development gives them flexibility and space to prioritize more technical labor and custom projects.

- **Low barrier to entry:** With simple tools, interfaces, and drag-and-drop features, low-code development allows a broader swath of workers access to apps and coding projects, helping them improve development knowledge and skills, and promoting a strong learning culture.
- **Bring apps to market quickly:** Preexisting templates and tools help workers produce and launch apps quickly.
- **Cost effectiveness:** Relying on low-code templates and tools means businesses don't have to invest the time and resources required for more complicated, custom development.
- **Simplified updates and maintenance:** Rather than depending on custom development for new updates and routine upkeep, low-code platforms can manage this on behalf of the business.
- **Helps keep organizations secure:** Platforms like Microsoft Power Platform offer built-in security and help bridge independent applications into a single structure.
- **Valuable data and analytics:** Built-in tools and dashboards within low-code platforms can offer visual models and reports with useful, actionable insights.
- **Automate repetitive tasks:** In conjunction with custom development, IT workers can use low-code development to optimize their workloads.

Drawbacks of low-code development

- **Less customizable and scalable than custom development:** While low-code platforms offer a broad range of pre-existing templates, they can be limiting for more complex projects, and don't always offer the ability to scale along with a business's growth and changing needs.
- **Limited integrations with other systems:** Legacy platforms or systems might require a more complex integration solution.
- **Lack of template flexibility:** Working within the specified template means workers may have to create workarounds or shift expectations about an app's functionality.

Examples of Traditional software development:

Many large applications are built primarily through traditional software

development, including:

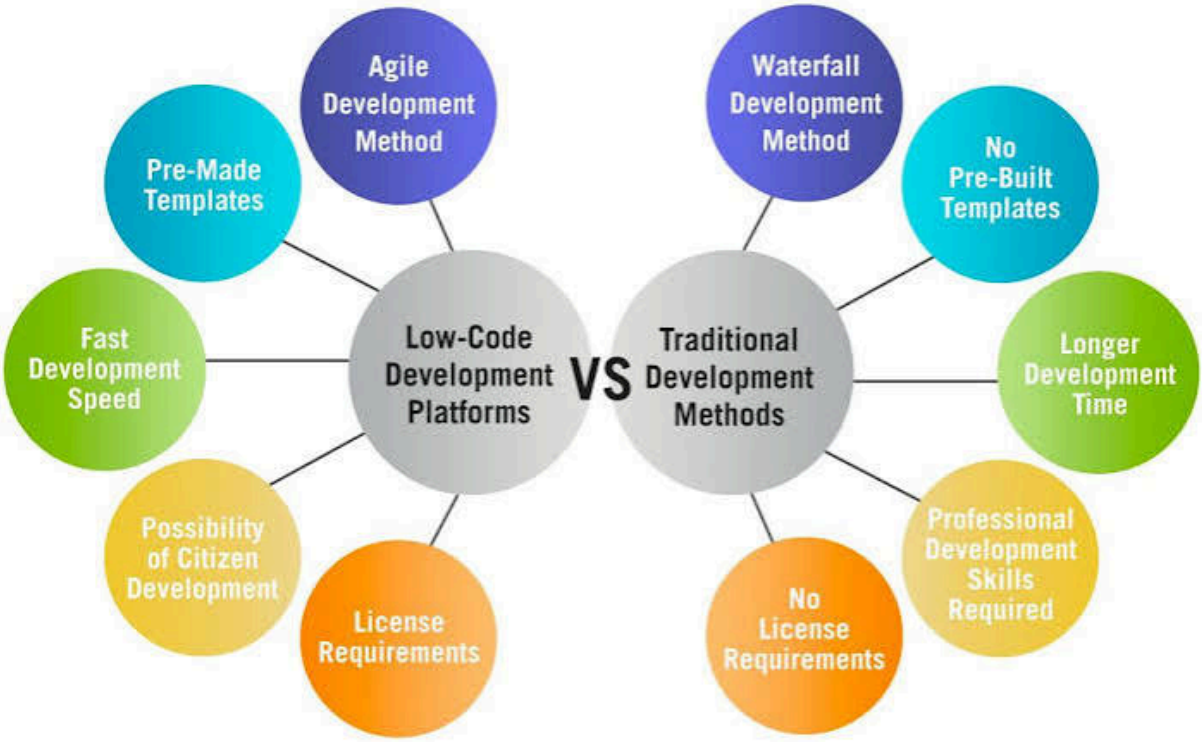
- Banking systems
- Airline reservation systems
- Hospital management systems
- Enterprise Resource Planning (ERP) software
- E-commerce platforms
- Social media platform

Benefits of custom app development

- **Highly customizable:** Custom developers can plan for, design, and meet specific business needs.
- **Scalable and adaptable:** Custom development is more agile, allowing IT professionals to pivot as necessary, scale up when required, and adapt to the changing needs of the business.
- **Expanded functionality:** Custom code offers a broad range of functionality, and developers can tailor functions to meet specific requirements.
- **Integration with other systems:** Custom developers can adapt the application to integrate with other legacy apps, systems, or products.

Drawbacks of custom app development

- **Continuous maintenance:** Custom apps require a more hands-on approach for security and upkeep.
- **Higher expenses:** A custom initial build can be costly and time consuming, and maintaining the custom app also requires specialized skills that can result in a higher bill.
- **Longer implementation and testing period:** As custom development is often complex, it can be more complicated to implement and test, resulting in a longer waiting period before releasing the app to customers or the public.
- **Updates and changes require more time:** Building a custom app involves a longer development process than using low-code development, and planning and launching updates takes more time as well.


Agile Development Method
Pre-Made Templates
Fast Development Speed
Low-Code Development Platforms
Possibility of Citizen Development
License Requirements
VS
Waterfall Development Method
No Pre-Built Templates
Longer Development Time
Traditional Development Methods
Professional Development Skills Required
No License Requirements